

**Before Sapiens:
The Hominin Field Before
the Surviving Species**

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Abstract

The taxonomic boundary around *Homo sapiens* falls late in the history of human capacities. Fossils from Jebel Irhoud place early members of the clade, or a population close to it, at about 315,000 years ago, and Omo I gives a minimum age of 233,000 years for eastern Africa. By then bipedal locomotion, stone-flake production, long-range dispersal, large-animal processing, Acheulean biface manufacture, fire use, wooden weapon production, and extended social learning already had long histories. Bipedality and stone working predate secure *Homo*. Range expansion, biface manufacture, fire practices, and organic technologies were sustained by *Homo erectus*, Neanderthal-lineage populations, Denisovans, *Homo naledi*, *Homo floresiensis*, and poorly resolved regional forms. We compile the earliest securely dated marker of eight capacities from the cited literature and compare each with the sapiens boundary. Seven of the eight predate both Jebel Irhoud and Omo I; the exception, wooden weapons, is known from a single site where wood happened to survive. Bipedal locomotion precedes the Jebel Irhoud boundary by 4.085 million years, the eight onsets span 4.2 million years, and the median lead is 1.885 million years. At least five other hominin lineages are dated near the boundary. The capacities later attributed to *H. sapiens* were built across an older, multi-lineage hominin population, which the surviving species inherited and narrowed by outlasting, absorbing, or replacing the other lineages.

1. Introduction

The oldest fossils usually placed near the base of the *Homo sapiens* clade come from Africa in the late Middle Pleistocene. The Jebel Irhoud material in Morocco, re-described by Hublin et al. (2017) and dated by Richter et al. (2017), is approximately 315,000 years old. Omo I, from the Omo Kibish Formation in Ethiopia, has a revised minimum age of 233,000 years (Vidal et al. 2022). Herto, from the Middle Awash, is later, at around 160,000 years ago (White et al. 2003). These dates spread the origin of the species across several African populations and morphologies.

The taxonomic label is useful: it allows fossils to be compared, regions to be sequenced, and lineages to be named. It also invites a compression of history, because the capacities later grouped under *H. sapiens* predate the point at which the label becomes stable. The species inherited a body plan older than the genus *Homo*, a technical history older than the earliest secure *Homo* fossils, and a social ecology shaped by cooperative foraging, infant dependency, and learned extraction of resources from difficult landscapes.

We therefore use “human” in a wider sense than the surviving species. The term here designates a field of lineages: the hominin bodies, techniques, social arrangements, and inherited niches through which *H. sapiens* later became possible. Differences between species remain real, but the modern species is not treated as the origin of capacities it inherited.

Before law, myth, ritual, or explicit doctrine can define persons, organisms must already be able to accumulate tools, food sharing, teaching, memory, and group inheritance. The archaeological and fossil record preserves this earlier assembly in fragments. Sections 2 to 9 review it capacity by capacity, and section 10 quantifies the chronology.

2. Bipedalism

The hominin line appears in the record through posture well before cranial expansion. *Sahelanthropus tchadensis*, described from Chad and dated to about 7 million years ago (Brunet et al. 2002), remains contested, but the debate already concerns the position of the foramen magnum and the possibility of upright locomotion. *Ardipithecus ramidus*, at 4.4 million years ago, provides a clearer body, capable of bipedal movement on the ground and still strongly adapted to arboreal life (White et al. 2009). The starting condition was a mosaic animal in a mosaic habitat.

By the time of *Australopithecus afarensis*, between roughly 3.9 and 2.9 million years ago, bipedality was an established part of the hominin repertoire. The Laetoli footprints, dated to 3.66 million years ago, preserve upright walking in volcanic ash (Leakey and Hay 1979). Lucy, at approximately 3.2 million years ago, shows the same through pelvis, knee, and lower limb. These hominins walked upright with brains far smaller than those of modern humans.

Bipedality changed more than gait. Upright posture freed the hands from routine weight-bearing and allowed infants, food, stone, wood, and carcass parts to be carried over distance; it exposed the face and chest for social display and changed the thermal load of daytime movement. A bipedal ape can bring material to a place and take material away, so that resources collected in one location can be used later in another.

The sequence matters. The upright body precedes large brains by millions of years, and the anatomical basis of human history lies first in the feet, pelvis, and hands and in habitat, long before cortical expansion.

3. Stone tools before secure *Homo*

The oldest widely discussed sharp-edged stone tools come from Lomekwi 3 in West Turkana, Kenya, dated to 3.3 million years ago (Harmand et al. 2015). They precede the earliest secure *Homo* fossils by several hundred thousand years, and their makers are unknown. Stone-flake production therefore begins without any secure association with the genus that later took the name human.

Oldowan technology appears later and spreads more widely. Gona in Ethiopia yields Oldowan assemblages from about 2.6 million years ago (Semaw et al. 1997). Nyayanga in Kenya, dated to between approximately 3.0 and 2.6 million years ago, associates Oldowan artifacts with hippopotamus butchery and *Paranthropus* teeth (Plummer et al. 2023). Early stone technology thus had several possible makers.

A flake is a produced edge, obtained by controlling angle, force, material, and sequence. The knapper must anticipate a cutting surface inside an unstruck core. The skill can be learned, repeated, and transmitted, so that even a simple tool rests on a tradition longer than any single act of manufacture, maintained across individuals.

Stone working is an early, minimal, and durable form of technical inheritance held partly outside the body. One individual learns from another, and each generation leaves cores, flakes, preferred

raw materials, activity sites, and failed attempts that structure the learning environment of the next. Technique becomes a social memory stored partly outside any single nervous system.

The archaeological object records the interaction of hand, material, food, landscape, and teaching. A small-brained hominin with a stable flaking tradition already lived in an inherited technical environment.

4. Early *Homo* and continental dispersal

The genus *Homo* enters the record in a taxonomically unsettled state. Fossils assigned to *Homo habilis*, *Homo rudolfensis*, early African *Homo erectus* or *Homo ergaster*, and other debated forms group differently depending on whether jaw, tooth, cranial, or postcranial traits are weighted. Brain size and body size increase unevenly. Species names mark clusters in a fragmentary record and do not describe a linear progression.

By roughly 1.8 million years ago early *Homo* populations had left Africa and reached Dmanisi in Georgia (Lordkipanidze et al. 2013). The Dmanisi fossils combine small brains, variable skull form, and stone tools, and their location shows that hominins had entered western Eurasia before the later human suite of traits existed.

Homo erectus is the longest-lasting lineage in this record. African and Eurasian populations assigned to *H. erectus* or closely related forms persisted over a very long span of time and space. The Nariokotome skeleton, about 1.5 million years old, has long legs and a body plan suited to covering distance (Brown et al. 1985). The species complex reached Java, China, the Caucasus, and Africa. For most of the period in which hominins were human in the wider sense used here, a form similar to *H. erectus* was the principal representative.

Acheulean technology, present by 1.76 million years ago at Kokiselei in Kenya (Lepre et al. 2011), is the most visible artifact of this expansion. The handaxe is a portable symmetrical form reproduced across large distances and long periods. Its uses varied, and its stability is more informative than any single function. Making a biface requires maintaining a planned shape through a sequence of removals, so the tool records planning carried out by subtraction.

Continental dispersal was a technical and social achievement. Moving across Africa and Eurasia meant making unfamiliar prey, raw materials, seasons, predators, parasites, and climates usable. Anatomy contributed, and learned ecological knowledge made the expanded range durable. A group needed to know where water could be found, which stone would fracture well, how carcasses could be opened, when travel was possible, how infants could survive it, and how to recognise danger. Hominins before *H. sapiens* already learned geography.

5. Fire use and maintenance

Fire is the hardest major capacity to date, because its traces are fragile and natural burning must be distinguished from controlled use. At Wonderwerk Cave in South Africa, Berna et al. (2012) report in situ burning inside the cave at roughly 1 million years ago. The inference of controlled use rests

on the microstratigraphy: burned bone and ashed plant matter lie in undisturbed layers deep inside the cave, beyond the reach of wildfire and lightning. Several European and Near Eastern sites provide later and stronger evidence of repeated fire use, and by the later Middle Pleistocene fire was a secure part of hominin life in many regions.

The presence of fire and its control must be kept separate. Burned bone alone is weak evidence for hearth-centred domestic life, and a hearth alone is weak evidence that cooking was universal. The conservative claim is already substantial: before *Homo sapiens*, some hominin groups brought fire into occupied spaces, kept it, and used it in ways that changed food, space, and the use of time.

Fire extends usable hours after sunset, keeps predators away from sleeping groups, softens food, and hardens wooden tools. It provides a place where individuals gather after foraging, adding a second phase to the day that is less occupied by search and movement.

Fire also imposes continuing work. A flame must be fed, guarded, carried, restarted, or found again; fuel must be gathered, embers watched, and the hearth protected from rain. A group that depends on fire takes on a maintenance task that outlasts any single use.

Later societies embedded fire in cosmology, sacrifice, metallurgy, cooking, domestic life, and industry. The earlier fact is simpler: fire gives a mobile group a durable centre, around which sustained attention, teaching, food sharing, and conversation at night become easier.

6. Organic technology and teaching

Large prey changes the social problem. Cut-marked bones, marrow extraction, hunting lesions, and kill or scavenging sites seldom distinguish hunting from aggressive scavenging cleanly, and the distinction matters less than the coordination either requires. Meat from a large animal is dangerous to obtain, heavy to move, divisible among many individuals, and perishable, so a carcass creates a short-lived economy of its own.

The Schöningen spears, redated to around 200,000 years ago (Hutson et al. 2025; originally described by Thieme 1997), show that hominins in Europe before *H. sapiens* made carefully shaped wooden weapons. The spears are usually attributed to archaic humans of the Neanderthal lineage or a closely related Middle Pleistocene population. Their preservation is exceptional, because wood rarely survives, and their existence implies a larger lost inventory of organic tools: digging sticks, carrying devices, handles, shafts, traps, and containers.

Such objects imply teaching. Complex wooden weapons require selection of raw material, shaping, balancing, and practice in use; stone knapping requires managed failure; fire requires maintenance. Children in these groups inherited capacities through more than genes. They grew up in an environment structured by adult practice and acquired it through demonstration, correction, imitation, tolerance, and play.

Extended childhood belongs to the same structure. Human infants are costly, slow to mature, and dependent, which is both an anatomical burden and a social opportunity. A long juvenile period gives time to acquire techniques that instinctive maturation alone would not supply. Cooperative

breeding, food sharing, and alloparental care allow such expensive learners to survive long enough to transmit a tradition.

Social transmission is visible even where symbols are absent. Teaching produces traditions more stable than independent individual discovery would, whether or not it leaves any mark resembling language. A handaxe industry, a fire practice, a hunting weapon, and a repeated butchery sequence are all records of social transmission.

7. Contemporary hominin lineages

Homo sapiens emerged into a world already occupied by other humans. Neanderthal-lineage populations were established in Europe and western Asia. Sima de los Huesos in Spain, dated to around 430,000 years ago, has yielded a large Middle Pleistocene assemblage that is genetically closer to Neanderthals than to Denisovans (Meyer et al. 2016). Denisovans, first identified from DNA in a small bone from Denisova Cave (Reich et al. 2010), had a population history in eastern Eurasia that fossil morphology had scarcely registered.

Other lineages complicate the picture further. *Homo naledi*, from Rising Star Cave in South Africa, is dated to between 335,000 and 236,000 years ago (Dirks et al. 2017) and was thus contemporary with early *H. sapiens* in Africa despite its small brain and unexpected morphology. *Homo floresiensis* survived on Flores until roughly 50,000 years ago, with roots that may extend much deeper into the history of island Southeast Asia. *Homo luzonensis*, described from Callao Cave in the Philippines and dated to at least 67,000 years ago for one element (Détroit et al. 2019), adds another island lineage with a mosaic of traits.

Ancient DNA changed the understanding of this period. Green et al. (2010) showed that Neanderthals contributed DNA to the ancestors of present-day non-African populations, amounting to a few percent of the genome in living people outside Africa. Reich et al. (2010) showed that Denisovans were a distinct archaic group whose ancestry survives especially in parts of Oceania and Asia. Later work identified further episodes of introgression among Neanderthals, Denisovans, and modern humans. The human tree in this period is reticulate, with branches that sometimes rejoin.

Species boundaries in the hominin record are sharp enough for taxonomy and permeable enough for ancestry. Anatomical difference, cultural difference, and reproductive separation coincided only imperfectly. Some groups diverged, adapted locally, met again, exchanged genes, and disappeared as recognisable populations while leaving traces in their descendants.

Restricting the word “human” to *Homo sapiens* obscures this structure. Neanderthals made tools, used pigments, cared for injured individuals, lived in harsh climates, and contributed to living genomes. Denisovans are known from genes before any well-described skeleton. *H. naledi* shows that morphology alone cannot explain behavioural capacity, since a small-brained hominin dated close to early *H. sapiens* may still have been capable of complex behaviour. A historical definition of the category has to precede any moral one.

8. The emergence of *Homo sapiens* in Africa

The emergence of *Homo sapiens* was a population process that extended across Africa. Jebel Irhoud combines a modern-looking face with other archaic features, and Omo I and Herto lie in eastern Africa. Genetic and archaeological evidence increasingly favours structured African populations linked by intermittent gene flow, ecological change, and regional cultural variation. Scerri et al. (2018) describe this as an origin in subdivided populations, with the explanatory weight placed on population structure across regions.

The transition to *sapiens* brought older capacities together under a new population regime. Upright bodies, manipulative hands, stone working, fire, cooperative foraging, teaching, and long-distance ranging were inherited. What changed was the density and flexibility with which they were recombined. The African Middle Stone Age shows regional technologies, prepared-core methods, hafting, pigments, beads, long-distance movement of materials, and broad ecological range, so that older capacities could now be combined in more varied ways.

The concept of “behavioural modernity” has often been asked to explain too much, because it implies a single threshold after which humans became fully human. The record is patchier. Pigment use appears in different places at different times; ornaments appear, disappear, and reappear; burial and mortuary treatment vary; long-distance exchange is intermittent; and complex technologies can be lost when populations shrink or ecologies change. Capacity depends on demography and social infrastructure as well as on cranial anatomy.

The emergence of *sapiens* is best described as the stabilisation of a package. The species that eventually spread across the planet inherited upright walking, stone working, range expansion, fire, cooperative hunting, and several learned traditions from older hominin populations, and its advantage lay in recombining them across larger, denser, and better-connected populations.

9. Single-variable origin accounts

Origin accounts commonly select one real variable and give it priority over all others. Encephalization is real, and brain size changed greatly across hominin evolution, but the chronology rules out a brain-first explanation: upright walking, manipulative hands, stone technology, and range expansion were all established before modern cranial form. Intelligence, as it appears in the record, is distributed across anatomy, group practice, landscape, and artifacts.

Species labels are indispensable and can mislead. They make comparison possible, but they encourage attributing to the last surviving species the histories of extinct ones. Any list of traits assigned to *Homo sapiens* usually contains older elements. The modern species holds and recombines inherited capacities, and treating each capacity as a *sapiens* invention compresses millions of years of hominin history into the most recent taxonomic label.

Technical origins raise the same problem at a different scale. A tradition is larger than an invention: stone industries, fire practices, hunting systems, and movement ecologies persist because groups

reproduce them. Inventions occur, but their survival depends on transmission, so a tool form that lasts for hundreds of thousands of years is a social achievement before it is an individual one.

Replacement was the outcome of a long process. *H. sapiens* became the only surviving hominin species and replaced other groups in many places, but the genetic and archaeological record preserves mixture, overlap, borrowing, coexistence, and local disappearance. The diversity of human lineages narrowed late.

Later cultural categories rest on this older foundation of bodies that carry, hands that make, groups that teach, food that is shared, fire that is maintained, and landscapes remembered across generations.

10. Quantified stratigraphy

The chronological claim can be stated numerically. We consider eight capacities with a single securely dated first appearance in the record reviewed above: bipedal locomotion, stone-flake production, large-animal butchery, Oldowan technology, continental dispersal, the Acheulean biface, controlled fire, and wooden weapons. Capacities without one defensible onset date, among them symbolic marking and habitual cooking, are excluded. For ranges, the midpoint is used (Nyayanga, 3.0 to 2.6 million years, enters as 2.8). Taking the earliest securely dated marker of each capacity, and the Jebel Irhoud date of 315,000 years as the sapiens boundary, seven of the eight capacities predate the boundary. Against the Omo I minimum of 233,000 years, the count is again seven of eight. The exception is wooden weapon manufacture, whose Schöningen instance was redated in 2025 to around 200,000 years (Hutson et al. 2025), after both Jebel Irhoud and Omo I; the late date reflects the rarity of preserved wood and is weak evidence for a late onset.

The leads are large. Bipedal locomotion, the deepest capacity in the set, precedes the boundary by 4.085 million years. The eight onsets span 4.2 million years from upright walking to shaped wooden spears, and the median lead over the boundary is 1.885 million years. Stone-flake production at Lomekwi precedes the earliest secure *Homo*, and continental dispersal at Dmanisi precedes the boundary by close to 1.5 million years. The ordering matches the sequence described in sections 2 to 6: upright bodies, then worked stone, then large-animal butchery, then dispersal, then fire, then the species boundary, with the single securely dated wooden weapon falling later because wood seldom survives.

At least five other hominin lineages are dated near the boundary, from the Neanderthal-lineage population at Sima de los Huesos at about 430,000 years through *Homo naledi*, the Denisovans, *Homo luzonensis*, and *Homo floresiensis*. The sapiens threshold therefore stabilised within an already populated field.

The dates are point estimates taken from the cited literature; no new measurements are reported. The computation reports only their stratigraphic relations.

11. Conclusion

The pre-sapiens record shows a layered inheritance. Hominins walked and carried; they learned to work stone; groups divided carcasses, maintained fire, moved through remembered landscapes, shaped wood, supported dependent learners, and kept fragile traditions alive across generations.

By the time *Homo sapiens* appears, this field of capacities was already old. Several lineages had built parts of it, leaving fossils, tools, DNA, or only gaps in the record. The modern species appears late in the sequence. It inherited capacities developed in other bodies and environments and became the population in which they were amplified, recombined, and extended after the other human lineages disappeared.

The earlier lineages took part in the same long history of technical, social, and ecological experimentation, and the surviving species is one outcome of it. Its distinctive features are real, and most of the capacities it is credited with are older than the species.

Reproducibility

The capability onsets, boundary dates, and contemporary lineages are listed with their sources in `simulation/stratigraphy.py`. Running `uv run run_all.py` in `simulation/` computes the lead times, counts, span, and median and writes them to `simulation/output/results.json`, together with a timeline figure in `simulation/output/figures/`.

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